A Multi-Level Label-Aware Semi-Supervised Framework for Remote Sensing Scene Classification

Qiang Liu, Min He, Yang Kuang, Linshan Wu, *Graduate Student Member, IEEE*, Jun Yue, and Leyuan Fang, *Senior Member, IEEE*

***Abstract*— Semi-supervised learning (SSL) is a promising approach to reduce the labeling burden in remote sensing scene classification tasks. However, most semi-supervised methods typically exploit the single-level semantic information of unlabeled data, ignoring the multi-level semantic structure prevalent in remote sensing data. The multi-level semantic structure, which contains the correlation of different categories and the multi-granularity semantic information, can help the scene classification model to more accurately measure the feature distance between different categories and more effectively utilize unlabeled data. Therefore, this article proposes a multi-level label-aware (MLLA) semi-supervised scene classification framework, MLLA, which extends the semantic information captured in unlabeled data from single-level to multi-level to improve the scene classification performance. Specifically, we first propose a multi-level prototype awareness module to capture the multi-level semantic structure underlying remote sensing data. Then, based on this structure, a multi-level pseudo-label generation module is designed to assign multi-level pseudo-labels to the unlabeled data. Finally, by combining the labeled samples and the multi-level pseudo-labeled samples, the scene classification model is progressively trained. The experimental results on three benchmark datasets show that the proposed MLLA achieves excellent performance compared to other semi-supervised classification methods.**

***Index Terms*— Deep learning, multi-level pseudo label, remote sensing, scene classification, semi-supervised learning (SSL).**

I. Introduction

REMOTE sensing satellites can dynamically and sustainably acquire a large number of remote sensing images containing detailed information about the Earth's surface. Remote sensing scene classification plays a critical role in many fields, such as urban planning [1], [2], [3], precision agriculture [4], [5], [6], and natural disaster detection [7], [8], [9]. With the rapid development of space technology and satellite sensor technology, the volume of remote sensing images is increasing dramatically. Manual-based remote sensing scene classification is labor-intensive and time-consuming, which limits its application in practice. Therefore, it is of great significance to develop a high-precision intelligent scene classification method.

Over the past few decades, many automatic remote sensing scene classification methods have been proposed [10], [11], [12], [13], [14], which can be broadly divided into two main categories: handcrafted feature-based methods and deep feature-based methods. Handcrafted feature-based methods distinguish remote sensing scenes based on low-level feature descriptors designed by human experts, such as texture descriptors [15], [16], color histograms [17], and histogram of oriented gradients [18]. However, the reliance on expert knowledge and the lack of description of higher order semantic features limit the performance and potential of these methods [19]. Recently, deep feature-based methods have attracted great attention, which can automatically capture high-level semantic features to classify remote sensing scenes through deep neural networks [20], [21], [22]. While deep feature-based methods have achieved excellent performance, they require a large amount of labeled data to train deep neural networks [23], [24], [25], [26], [27], [28], [29]. The high cost of data annotation limits the application of deep feature-based methods in real-world scenarios.

Semi-supervised learning (SSL) is an effective paradigm for leveraging unlabeled data to alleviate the burden of data annotation [30], [31], [32], [33], [34], [36]. Recently, some semi-supervised methods have been proposed. For example, Lee et al. [37] proposed a simple and effective semi-supervised classification method, pseudo label, which took the class with the highest prediction probability of unlabeled data as the pseudo-label. The model was then trained by combining labeled and pseudo-labeled data. Xie et al. [38] trained a classification model by applying strong data augmentation to unlabeled data, and enforcing consistency in the model's predictions across multiple different augmented versions. Sohn et al. [39] proposed Fixmatch, which trained a classification model using pseudo-labels generated from the weakly augmented predictions of unlabeled data. The aforementioned method effectively improves classification accuracy by utilizing the semantic information contained in unlabeled samples.

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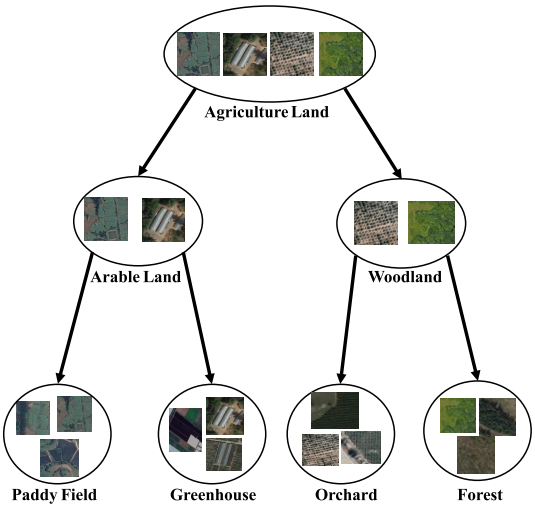


Fig. 1. Schematic of multi-level semantic structure in remote sensing images. For example, remote sensing images of paddy fields and greenhouses can be classified as arable land at a higher semantic level, and orchards and forests can be grouped into the category of woodland. Finally, both arable land and woodland can be classified as agriculture land.

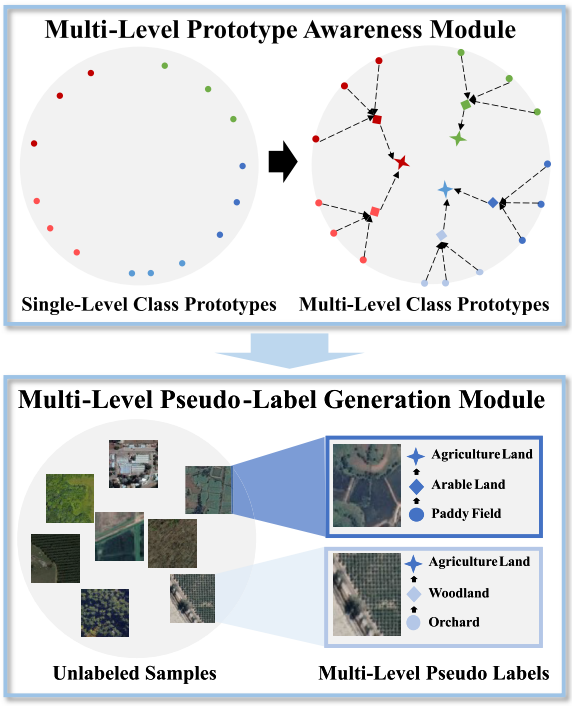


Fig. 2. Schematic illustration of the proposed method: First, we model the multi-level semantic structure in remote sensing data via the multi-level prototype awareness module. Then, based on this structure, the multi-level pseudo-label generation module assigns multi-level semantic pseudo-labels to unlabeled samples for training the scene classification model.

However, existing semi-supervised methods typically learn single-level semantic information from unlabeled images, without considering the multi-level semantic structure generally presented in remote sensing images [40], [41], [42], [43]. As shown in Fig. 1, remote sensing images exhibit a multi-level semantic structure, in which both forest and orchard can be classified as woodland, and their correlation with each other is stronger than their correlation with arable land. This semantic correlation can help the classification model to more accurately measure the distance between different remote sensing scene categories in the feature space. In addition, coarsely classifying a remote sensing image as woodland or arable land is less challenging than identifying it as orchard and forest in fine detail. Therefore, the unlabeled data, which is difficult to be analyzed at a fine-grained level, can be utilized effectively by coarse-grained analysis.

Inspired by the above-mentioned multi-level semantic structures, we propose a multi-level label-aware (MLLA) semi-supervised framework for remote sensing scene classification, called MLLA, which can capture the multi-level semantic information in unlabeled data to improve scene classification performance. As shown in Fig. 2, we first propose a multi-level prototype awareness module to model the multi-level semantic structure in remote sensing data. Furthermore, a multi-level pseudo-label generation module is designed to assign multi-level semantic pseudo-labels to unlabeled data for training the scene classification model. The proposed method does not require any additional annotation, and can be applied to common single-level annotated remote sensing datasets.

The remainder of this article is organized as follows. Section II reviews the related work in this article. Section III describes the details of the proposed framework. The comprehensive experimental results are presented in Section IV. Sections V and VI discuss and conclude this work, respectively.

II. Related Works

In this section, we briefly introduce the existing work related to our method, including remote sensing scene classification and SSL.

A. Remote Sensing Scene Classification

In the past few years, deep learning technique has made a remarkable progress in the field of remote sensing scene classification due to its powerful feature extraction capability. Anwer et al. [44] proposed a two-stream deep fusion network for remote sensing scene classification. In this method, the color information of remote sensing image was combined with the texture information of map-encoded image to improve the performance of scene classification. Xie et al. [45] proposed a scale-free convolutional neural network, which could classify remote sensing images of arbitrary size to avoid the loss of key information caused by image resizing process. Xu et al. [46] developed a deep feature aggregation framework driven by graph convolutional network. In the framework, the context information between the deep features extracted from remote sensing images was mined for better scene classification. Ma et al. [47] designed a multi-objective neural evolutionary search algorithm for scene classification network structure. In this method, a competitive neural structure was obtained by jointly optimizing the test error and computational complexity of the searched network. Chen et al. [22] proposed a multi-branch local attention network, which obtained both channel and spatial attention to emphasize the main target in the complex background of remote sensing images. One major

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factor in the success of the deep learning-based methods is the large amount of labeled data used to train the network. However, labeling large amounts of remote sensing data is labor-intensive and time-consuming, which limits the practical application of the above scene classification methods.

B. Semi-Supervised Learning

SSL is an effective paradigm to exploit the information in unlabeled data to improve model performance. A considerable number of studies have been conducted on SSL, where consistency regularization and pseudo-labeling are two popular categories of methods [30], [48], [49].

Consistency regularization [50] enforces the model to output the same predictions for unlabeled images with different perturbations. A variety of data perturbations have been applied to the consistency regularization-based methods. For instance, Laine et al. [51] proposed the Π-model, which applied different random perturbations to the same unlabeled image and enforced the model predictions to be consistent with a consistency loss. Miyato et al. [52] first introduced adversarial perturbations in SSL, via the adversarial training to improve the model performance. Berthelot et al. [53] mixed labeled and unlabeled data, and further enforced the model outputs of the mixed data to be consistent with the mixed label of labeled data and unlabeled data. Xie et al. [38] employed stronger data augmentation strategy, RandAugment [54], to impose stronger perturbations on unlabeled data.

Pseudo-labeling methods [37], [55] generate pseudo-labels for some unlabeled samples predicted with high confidence, and then learns from the pseudo-labeled data. Lee et al. [37] picked the category with the maximum predicted probability as the pseudo-labels for the unlabeled data. However, the pseudo-labeling methods suffer from confidence bias when the pseudo-labels are inaccurate. To ensure the reliability of pseudo-labels, Chen et al. [56] trained three classifiers to generate pseudo-labels for unlabeled data if two of them have consistent predictions and the prediction confidence is over a threshold. Recently, some hybrid approaches that integrate consistent regularization and pseudo-labeling into a unified framework have achieved significant performance gains. For example, unsupervised data augmentation (UDA) [38] selected high confidence predictions derived from unlabeled samples as soft labels. Then, the predictions of strong augmented data were enforced to be consistent with the soft labels. FixMatch [39] generated hard labels from unlabeled data with weak noise, and then trained the model using the corresponding unlabeled data with strong noise. Although the above methods achieve excellent performance improvement, the multi-level semantic information in remote sensing images is ignored. Different from the above methods, we aim to extend the semantic information captured in remote sensing unlabeled data from single-level to multi-level, which can help the model measure the distance of different scene categories more accurately in feature space and utilize unlabeled data more efficiently.

III. Methodology

In this section, we first present a brief review of the related background in Section III-A. Second, we introduce the overall framework of the proposed MLLA in Section III-B. Then, two core modules of MLLA, i.e., the multi-level prototype awareness module and the multi-level pseudo-label generation module, are introduced in detail in Sections III-C and III-D, respectively. Finally, we described the training loss of MLLA.

A. Background

In the setting of a semi-supervised remote sensing scene classification task, let $X = \{(x_b, y_b)\}_{b=1}^{B}$ be a batch of B labeled data, where x_b is a remote sensing image and y_b denotes the label of the image. Let $U = \{u_b\}_{b=1}^{\mu B}$ be a batch of μB unlabeled data, where μ determines the ratio of X and U. By combining X and U, the parameters θ of the scene classification model $f_\theta(\cdot)$ are optimized.

Fixmatch is a typical SSL method related to our method, which is briefly described below. In the training phase, for a batch of labeled remote sensing samples X, FixMatch calculates the supervised loss L_x on weakly augmented labeled samples in (1). For a batch of unlabeled remote sensing samples U, two branches are designed. In one branch, FixMatch takes the weakly augmented version $\mathrm{Aug}_w(U)$ into the classification model and obtains predictions $P = f_\theta(\mathrm{Aug}_w(U))$. Then, an arg max operation is performed on the predictions to obtain pseudo labels $\arg\max(P)$. Furthermore, FixMatch filters out a large amount of noisy pseudo-labels by a high threshold τ. In the other branch, a strongly augmentation is applied to U, then the unsupervised loss L_u between the predictions of the strongly augmented version $\mathrm{Aug}_s(U)$ and the pseudo labels is calculated in (2). The goal of FixMatch is to minimize the total loss L_{fixmatch}, which can be defined in (3)

$$L_x = \frac{1}{B}\sum_{b=1}^{B} H\left(y_b, f\left(\mathrm{Aug}_w(x_b)\right)\right) \tag{1}$$

where $H(q, p)$ is the cross-entropy between distributions q and p, $\mathrm{Aug}_w(\cdot)$ is a weak data augmentation function

$$L_u = \frac{1}{\mu B}\sum_{b=1}^{\mu B} I(\max(p_b) \geq \tau) H\left(q_b, f_\theta\left(\mathrm{Aug}_s(u_b)\right)\right) \tag{2}$$

where $\mathrm{Aug}_s(\cdot)$ is a strong data augmentation function, $p_b \in P$, $q_b = \arg\max(p_b)$, and $I(\cdot)$ is an indicator function to filter out noisy pseudo-labels whose largest class probability fell below the threshold τ

$$L_{\text{fixmatch}} = L_x + \lambda L_u \tag{3}$$

where λ is a weight coefficient determining the weight of the unsupervised loss L_u.

However, Fixmath ignores the multi-level semantic information and assigns only a single-level pseudo-label to each unlabeled sample, which is inefficient and difficult to model the semantic relations of remote sensing images.

B. Overall Framework

The overall framework of MLLA is illustrated in Fig. 3, in which the classification model consists of two learnable parts: 1) a feature extractor $g(\cdot)$ and 2) a multi-head

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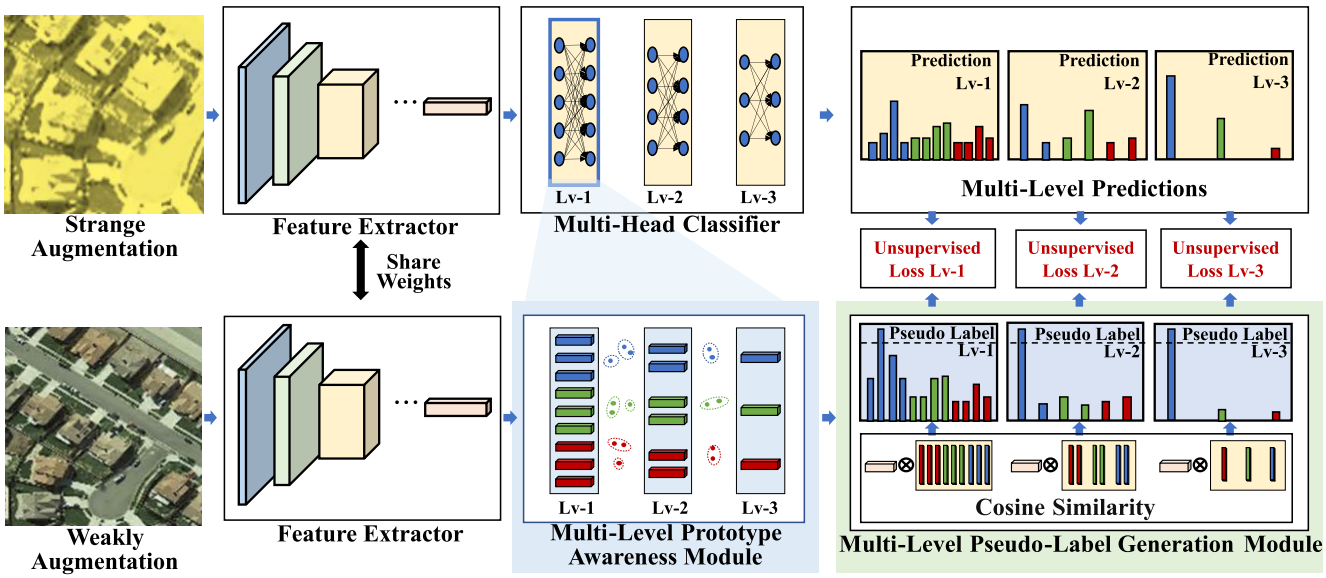


Fig. 3. Overview of MLLA framework, which contains two core modules: a multi-level prototype awareness module and a multi-level pseudo-label generation module. First, we model the multi-level semantic structure in remote sensing images via the multi-level prototypes awareness module. Then, based on the multi-level semantic structure, we design a multi-level pseudo-label generation module to assign multi-level pseudo-labels for unlabeled remote sensing samples. Finally, by combining the labeled samples and multi-level pseudo-labeled samples, the feature extractor and the multi-head classifier are trained progressively.

classifier $\{h_l(\cdot)\}_{l=1}^{L}$, where L denotes the semantic level of the prediction. Different from most of existing semi-supervised methods [39], [57], [58] that only exploit single-level pseudo labels, our proposed MLLA extends pseudo-labels from single-level to multi-level to train the classification model. Specifically, we design two core modules: a multi-level prototype awareness module and a multi-level pseudo-label generation module. First, we model the multi-level semantic structure in remote sensing images via the multi-level prototype awareness module. Then, based on the multi-level semantic structure, we design the multi-level pseudo-label generation module to assign multi-level pseudo-labels for unlabeled remote sensing samples. Finally, by combining the labeled samples and the multi-level pseudo-labeled samples, the feature extractor $g(\cdot)$ and the multi-head classifier $\{h_l(\cdot)\}_{l=1}^{L}$ are trained progressively. Each step of the proposed method will be elaborated in the following.

C. Multi-Level Prototype Awareness Module

The core of our approach is to generate multi-level pseudo-labels for unlabeled data by the multi-level semantic structure in remote sensing data. However, in real applications, most remote sensing datasets provide only single-level semantic annotation, which cannot directly capture the multi-level semantic structure. To solve this issue, we propose the multi-level prototype awareness module to model the multi-level semantic structure in remote sensing data.

Algorithm 1 gives details about the multi-level prototype awareness module. In MLLA's multi-head classifier $\{h_l(\cdot)\}_{l=1}^{L}$, the level 1 classification head $h_1(\cdot)$ is a fully connected layer, parameterized by the weight matrix $W \in \mathbb{R}^{(D\times N_1)}$. Here, the parameter matrix W denotes a collection of class vectors $[w_1, \ldots, w_{N_1}]$, w_{N_1} is a D-dimensional class vector of the N_1th class. We perform the L_2 normalization operation on the class vector to obtain the first level class prototypes $\{c_1^i\}_{i=1}^{N_1}$, where $c_1^i = w_i/\|w_i\|$. Subsequently, the higher level semantic prototypes are obtained by performing k-means clustering [59] to the lower level prototypes. Finally, a set of multi-level prototypes $C = \{\{c_l^i\}_{i=1}^{N_l}\}_l^L$ are formed, consisting of L layers, each with N_l prototypes. In this set of multi-level prototypes, higher- level prototypes $\{c_l^i\}_{i=1}^{N_l}$ can be considered as parents of lower level prototype $\{c_{l-1}^i\}_{i=1}^{N_{l-1}}$ and are easier to identified. It is worth noting that the multi-level prototype awareness module requires no additional annotation, and can be used for common single-level annotated remote sensing datasets.

As the training process progresses, the weight vectors of the classifier, i.e., the first-level prototypes, are updated. To represent the multi-level semantic structure in remote sensing images more accurately, the multi-level prototypes require continuous iterations. In our method, we perform multi-level clustering every one batch to ensure the timely update of the multi-level prototypes. The multi-level clustering is highly efficient, since it is performed only on the class prototypes.

Algorithm 1 Multi-Level Prototype Awareness Module

Given: the weight matrix $W = [w_1, \ldots, w_{N_1}] \in \mathbb{R}^{D\times N_1}$ of level 1 classification head $h_1(\cdot)$.
Output: multi-level prototypes $C = \{\{c_l^i\}_{i=1}^{N_l}\}_l^L$
$\{c_1^i\}_{i=1}^{N_1} \leftarrow \{w_i/\|w_i\|\}_{i=1}^{N_1}$
for l=2 to L **do**
  $\{c_l^i\}_{i=1}^{N_l} \leftarrow$ K-means$(\{c_{l-1}^i\}_{i=1}^{N_{l-1}})$
end for

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D. Multi-Level Pseudo-Label Generation Module

Existing semi-supervised methods typically filter out a large amount of low-confidence unlabeled data to prevent model performance from being disturbed. However, the ignored data still contains a lot of valuable information for model training. For example, an image may be indistinguishable as an orchard or forest, but can be classified as a woodland at a coarser granularity. Instead of just discarding it, producing a higher level pseudo-label might teach the model how to distinguish it from other categories such as arable land and bare ground. Inspired by this, we design a multi-level pseudo-label generation module to make full use of multi-level semantic information in remote sensing data.

For an unlabeled remote sensing image x, we first obtain the image representation $z = g(\mathrm{Aug}_w(x))$ by feeding the weak augmented view $\mathrm{Aug}_w(x)$ into the feature extractor $g(\cdot)$. Let $s(z, c) = z^T \cdot c/\|z\|$ notate the cosine similarity between the L_2 normalized z and the category prototype $c \in C$, By calculating the cosine similarity between feature z and the above multi-level prototypes $C = \{\{c_l^i\}_{i=1}^{N_l}\}_l^L$, we obtain the multi-level predicted probability $P_{\text{multi}} = \{\{p_l^i\}_{i=1}^{N_l}\}_l^L$, which can be formulated as follows:

$$p_l^i = \frac{\exp\left(s\left(z, c_l^i\right)/t\right)}{\sum_{k=1}^{N_l} \exp\left(s\left(z, c_l^k\right)/t\right)} \tag{4}$$

where N_l is the number of categories at l level and t denotes a temperature parameter.

Due to the possibility of incorrect predictions, using predicted results directly as pseudo-labels may degrade the model performance. To guarantee the quality of the pseudo labels, we pre-set a confidence threshold τ, and only generate multi-level pseudo labels for unlabeled samples with high prediction confidence. Concretely, for the prediction $p_l = \{p_l^i\}_i^{N_l}$ at level l, only if the largest predicted probability $\max(p_l) \geq \tau$, we use $q_l = \arg\max(p_l)$ as the pseudo label at level l. We repeat the above pseudo-label generation operation from level 1 to level L to get the multi-level pseudo labels $\{q_l\}_{l=1}^L$.

In addition, an intra-class feature mixture operation is applied for the multi-level pseudo-labeled samples. At a category level above the first level, we randomly select two samples x_a and x_b with the same pseudo-label. Then, we extract their deep features $z_a = g(x_a)$ and $z_b = g(x_b)$, respectively. The mixed deep feature z_{mix} is generated as follows:

$$z_{\text{mix}} = \alpha z_a + (1 - \alpha) z_b \tag{5}$$

where $\alpha \in [0, 1]$ controls the mixing ratio of feature z_a and z_b.

In the multi-level pseudo-label generation module, the samples that are difficult to distinguish in the sub-classes can be exploited by giving them more easily distinguishable parent labels. Compared with the previous single-level pseudo-labeling methods, our proposed MLLA can not only capture the multi-level semantic structure in remote sensing data, but also generate pseudo-labels for more unlabeled samples.

E. Loss Functions

In MLLA, we aim to optimize the feature extractor $g(\cdot)$ and the multi-head classifier $\{h_l(\cdot)\}_{l=1}^L$ by combining two loss terms: a supervised loss on labeled data L_x and a multi-level unsupervised loss L_{multi} on unlabeled data. Specifically, given a batch of B labeled samples $X = \{(x_b, y_b)\}_{b=1}^B$, we calculate the cross-entropy between the labels and the model's predictions of weak augmented views as the supervised loss L_x

$$L_x = \frac{1}{B} \sum_{b=1}^{B} H\left(y_b, h_1 \circ g\left(\mathrm{Aug}_w(x_b)\right)\right). \tag{6}$$

For a batch of unlabeled samples $U = \{u_b\}_{b=1}^{\mu B}$, the multi-level unsupervised loss L_{multi} is defined as the cross-entropy between multi-level pseudo-labels and model's predictions of strongly augmented unlabeled data

$$L_{\text{multi}} = \frac{1}{\mu B} \sum_{b=1}^{\mu B} \sum_{l=1}^{L} I\left(\max\left(p_l^b\right) \geq \tau\right) H\left(q_l^b, \hat{p}_l^b\right) \tag{7}$$

where $p_l^b = h_l \circ g(\mathrm{Aug}_w(u_b))$, $\hat{p}_l^b = h_l \circ g(\mathrm{Aug}_s(u_b))$ and $q_l^b = \arg\max(p_l^b)$. If the level L is set to 1, the multi-level loss would degenerate to the single-level unsupervised loss in (2). Thus, the single-level unsupervised loss can be regarded as a particular case of the multi-level unsupervised loss. The overall loss function is defined as following:

$$L_{\text{MLLA}} = L_x + \lambda L_{\text{multi}} \tag{8}$$

where λ is weight coefficient controlling the weight of the multi-level unsupervised loss.

It is worth noting that, in the training phase, all classification heads of the multi-head classifier are used to optimize the model, while in the testing phase, to reduce the computational cost, we keep only the classification head corresponding to the category level that needs to be classified.

IV. Experiments

To evaluate the performance of the proposed method, we conducted extensive experiments on three popular scene classification datasets. In this section, we first give a description of the datasets and detail the experiment setting. Then, we compare our method with the state-off-the-art SSL methods on the above three dataset. Finally, to investigate the effect of different factors on our method, we conduct ablation experiments.

A. Dataset Description

To evaluate our proposed method, three publicly remote sensing scene classification datasets are used in the experiments: the UC Merced dataset [60], the AID dataset [61], and the NWPU-RESISC45 dataset [12].

The UC Merced dataset is the first publicly available remote sensing scene classification dataset. This dataset contains 2100 aerial images grouped into 21 typical land-use classes, covering agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park,

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overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis court. Each classes contains 100 images with three RGB channels. Each image measures 256 × 256 pixels with a spatial resolution at 0.3 m.

The AID dataset is a large-scale remote sensing dataset acquired from Google Earth. It comprises 10 000 remote sensing images divided into 30 aerial scene categories: airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, and viaduct. Each category contains approximately 200–400 images in the RGB color space. The images are 600 × 600 pixels in size with a resolution range of 0.5–0.8 m.

The NWPU-RESISC45 dataset is one of the largest remote sensing scene classification datasets, which consists of 31 500 images with 45 scene classes: airplane, Pseudo-airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snow berg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland. Each class comprises 700 remote sensing images in RGB space. The size of each image is 224 × 224 pixels and the spatial resolution is ranged from 0.2–30 m.

B. Experiment Settings

1) Dataset Settings: In the experiments, each dataset is divided into a training set, a validation set and a test set in the proportion of 60%, 10%, and 30%, respectively. To evaluate the performance of our method with different numbers of labels, we select one, two, five, and ten samples per class from the training set to form labeled subsets with different numbers of labels, and use the rest of the data as unlabeled subsets.

2) Model Details: We adopt the ResNet34 [62] model as the feature extractor $g(\cdot)$, and initialize the feature extractor with pretrained weights from ImageNet-1K [63]. For the multi head classifier $\{h_l(\cdot)\}_{l=1}^{L}$, we set the number of classification heads L to 3. The main classification head $h_1(\cdot)$ at level 1 is a liner projection, and the other auxiliary classification heads are three-layer MLP projections. For each MLP projection, the first two layers consist of a fully connected layer, batch normalization, ReLU activation, and Dropout regularization with an output dimension of 1024, and the last layer is a simple fully connected layer for classification. To improve training stability, a L_2 normalization is applied to the input features and weight parameters of the last layer of all classifiers. When constructing the multi-level prototypes, the number of prototypes from first to third level are set to $\{c, \lfloor c/2 \rfloor, \lfloor c/4 \rfloor\}$, where c denotes the number of classes contained in a dataset. Specifically, the number of prototypes is set to 21, 10, 5 in the UC Merced dataset, to 30, 15, 7 in the AID dataset, and to 45, 22, 11 in the NWPU-RESISC45 dataset.

TABLE I
COMPARISON EXPERIMENT ON THE UC MERCED DATASET. RED INDICATES THE HIGHEST SCORE AND BLUE INDICATES THE SECOND HIGHEST SCORE

Method	Accuracy			
	1 labels	2 labels	5 labels	10 labels
Supervised	**44.57±5.46**	55.29±6.62	74.90±4.13	84.33±2.02
Pseudo-Label [37]	40.43±9.72	57.67±5.73	78.52±4.36	86.48±3.43
Mixmatch [53]	25.57±7.81	51.19±6.88	72.05±3.40	81.52±1.92
UDA [38]	40.86±1.74	61.57±3.87	77.90±3.10	86.24±2.15
VAT [52]	36.95±8.13	56.29±9.12	78.48±4.48	85.05±2.63
Fixmatch [39]	41.86±3.97	61.05±6.32	78.80±3.05	**87.57±2.03**
Flexmatch [64]	36.95±5.30	57.33±4.77	77.10±2.66	84.52±3.33
Dash [65]	38.52±4.64	55.57±7.43	73.33±3.83	82.52±3.56
Softmatch [66]	34.76±5.53	52.71±5.74	76.29±0.95	84.10±2.50
Freematch [59]	41.43±3.25	**62.48±7.85**	**79.10±1.68**	86.24±2.10
MLLA	**49.29±3.40**	**71.22±4.33**	**84.38±3.93**	**88.52±1.81**

3) Training Settings: The model is trained using the SGD optimizer with a weight decay rate of 0.0005 and a momentum of 0.9. The training epoch is 200, the batch size of labeled data is 16 and the batch size of unlabeled data is 224. During the training process, the learning rate increases linearly to 0.001 in the first five epochs and then gradually decays with a cosine scheduler. For the multi-level unsupervised loss, we set its weight coefficients to 1. The temperature parameter t and the confidence threshold τ for the pseudo-labels are set to 0.07 and 0.95, respectively. For the data augmentation strategies, we use RandAugment as the strong augmentation, and use random horizontal flip as the weak augmentation. All experiments are run on a server with 8 NVIDIA RTX 3090 GPUs and the entire framework is implemented using the PyTorch 1.9 library. We use the overall accuracy as the evaluation metric and report the mean and standard deviation of overall accuracy from five replicate experiments.

C. Experimental Results and Comparisons

Following the above experimental protocols, our method is compared with the state-of-the-art semi-supervised classification methods. We report the scene classification accuracies using different numbers of labeled data on the three aforementioned datasets.

1) UC Merced Dataset: Table I reports the comparison of the proposed MLLA with representative semi-supervised classification methods on the UC Merced dataset. MLLA achieves superior performance, with accuracies of 49.29%, 71.22%, 84.38%, and 88.52% for one, two, five, and ten labeled samples per class, respectively. It is worth noting that our method demonstrates significant improvement with fewer labeled samples per class. For instance, with only one or two labels per class, our method outperforms the second-best method by 4.72% and 8.74%, respectively. To illustrate the classification results for each specific category in detail, in Fig. 4, we show

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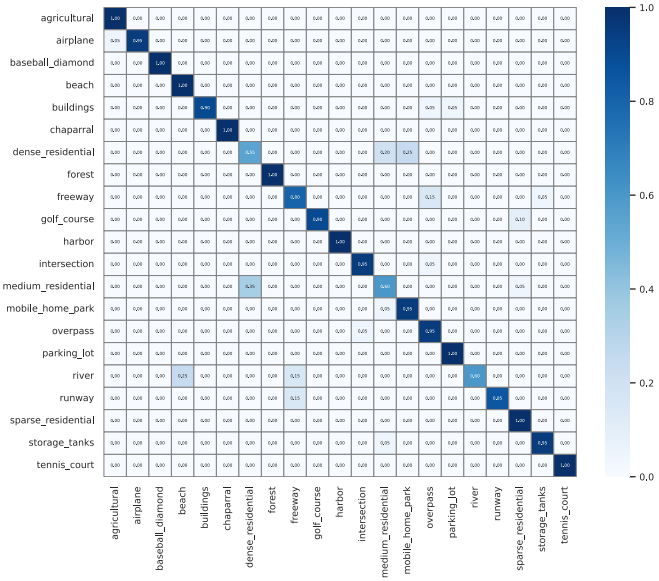


Fig. 4. Confusion matrix obtained by the proposed MLLA on the UC Merced dataset with ten labeled training samples per class.

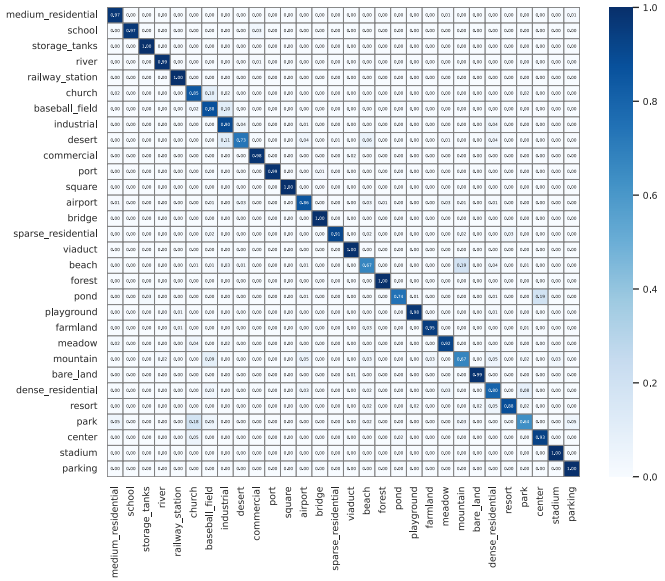


Fig. 5. Confusion matrix obtained by the proposed MLLA on the AID dataset with ten labeled training samples per class.

TABLE II

COMPARISON EXPERIMENT ON THE AID DATASET. RED INDICATES THE HIGHEST SCORE AND BLUE INDICATES THE SECOND HIGHEST SCORE

Method	Accuracy			
	1 labels	2 labels	5 labels	10 labels
Supervised	41.60±2.72	55.12±2.09	70.37±1.26	79.71±1.66
Pseudo-Label [37]	40.59±3.18	61.48±1.58	80.09±2.32	84.97±2.09
Mixmatch [53]	14.16±4.60	57.90±6.32	77.22±1.36	81.86±1.79
UDA [38]	46.19±4.76	67.94±6.13	**84.20±2.40**	**89.29±1.44**
VAT [52]	19.65±3.62	54.53±4.42	77.23±2.12	83.18±1.99
Fixmatch [39]	46.07±6.11	67.97±5.00	83.43±3.53	88.80±2.02
Flexmatch [64]	56.06±4.17	68.87±3.41	78.65±2.69	85.58±1.10
Dash [65]	45.29±3.17	61.48±4.19	76.06±3.18	84.43±0.74
Softmatch [66]	55.30±4.80	65.73±4.07	79.67±1.68	85.13±1.48
Freematch [59]	**58.93±3.95**	**70.77±2.90**	81.26±2.70	87.19±0.77
MLLA	**58.67±5.43**	**75.32±2.96**	**85.72±2.07**	**90.01±0.82**

TABLE III

COMPARISON EXPERIMENT ON THE NWPU-RESISC45 DATASET. RED INDICATES THE HIGHEST SCORE AND BLUE INDICATES THE SECOND HIGHEST SCORE

Method	Accuracy			
	1 labels	2 labels	5 labels	10 labels
Supervised	34.62±1.43	47.87±1.11	61.12±0.09	76.30±1.28
Pseudo-Label [37]	33.03±2.35	50.78±1.42	74.65±0.48	82.56±0.93
Mixmatch [53]	13.67±1.48	46.01±4.17	70.36±1.51	78.73±2.78
UDA [38]	34.62±5.10	61.62±2.41	82.55±1.60	87.93±0.84
VAT [52]	17.91±2.39	37.30±1.90	66.66±2.54	77.61±1.44
Fixmatch [39]	32.30±4.75	61.25±3.07	82.15±1.09	88.31±0.54
Flexmatch [64]	44.24±2.48	64.87±3.07	81.13±1.47	87.35±0.48
Dash [65]	33.47±5.35	56.17±2.33	77.78±2.53	85.42±0.92
Softmatch [66]	39.75±3.30	66.44±3.43	80.81±1.00	86.97±0.51
Freematch [59]	**46.90±6.40**	**68.24±3.04**	**83.57±1.14**	**88.79±0.68**
MLLA	**48.19±1.30**	**69.25±3.19**	**84.71±0.86**	**89.13±0.61**

the confusion matrix of the proposed MLLA with ten labeled training samples per class. Our MLLA achieves over 90% accuracy in 16 out of the 21 categories, despite having only ten labeled training samples per category. The misclassification mainly occurs between highly similar classes in remote sensing images, such as dense versus medium residential, rivers versus beaches, and highways versus runways. Experimental results demonstrate that the proposed MLLA achieves decent performance on the UC Merced dataset, even with limited labeled training samples.

2) AID Dataset: Table II shows the outperformance of the proposed MLLA on the AID dataset. MLLA is significantly superior to the previous methods. Specifically, MLLA achieves an accuracy of 58.67% with only one labeled training samples per class, which is comparable to the previous best-performing method, Freematch [57]. With two labeled training samples per class, MLLA outperforms the second-best method, Freematch, by 4.55%, reaching an accuracy of 75.32%. Given five labeled training samples per class, MLLA achieves an accuracy of 85.72%, a 1.52% improvement over UDA [38]. With ten labeled training samples per class, MLLA achieves an accuracy of 90.01%, which is 0.72% better than the previous best method, UDA. Fig. 5 shows the confusion matrix using the proposed MLLA on the AID dataset. As we can see, the accuracy of our method exceeds 90% in 20 of the 30 categories. The accuracy even reaches 100% in eight categories: storage tanks, railway station, square, bridge, viaduct, forest, stadium, and parking. The promising results show that it is feasible to classify the classes involved in the AID dataset using the proposed MLLA.

3) NWPU-RESISC45 Dataset: As listed in Table III, the proposed MLLA also achieves promising performance on the NWPU-RESISC45 dataset. Specifically, with one, two, five, and ten labeled training samples of each class, the proposed

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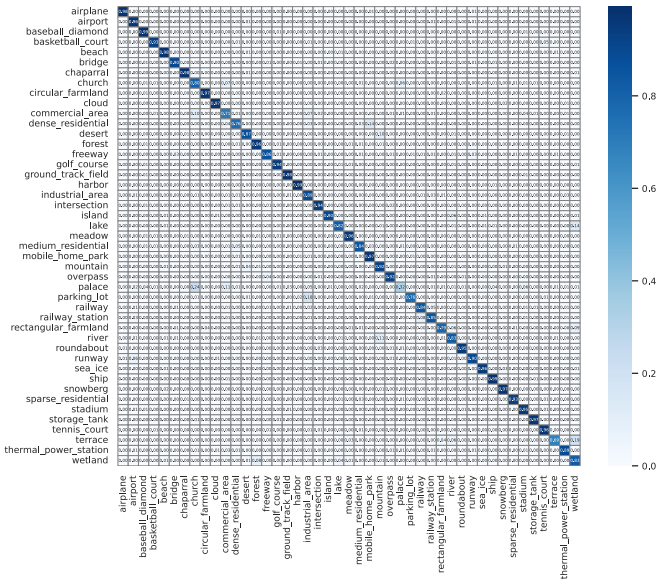


Fig. 6. Confusion matrix obtained by the proposed MLLA on the NWPU-RESISC45 dataset with ten labeled training samples per class.

MLLA obtains accuracies of 48.19%, 69.25%, 84.71%, and 89.13%, respectively. In comparison to the previous state-of-the-art method, Freematch, the proposed MLLA gains 1.29%, 1.01%, 1.14%, and 0.34% point accuracy improvement for one, two, five, and ten labeled training samples per class. Moreover, MLLA exhibits a lower standard deviation, particularly with only one labeled training sample per class, by 5.1% points compared to Freematch. This indicates that the proposed MLLA outperforms Freematch with fewer labeled samples. Fig. 6 depicts the confusion matrix of MLLA with ten labeled training samples per class on the NWPU-RESISC45 dataset. Despite the presence of numerous complex and challenging scene categories in the NWPU-RESISC45 dataset, MLLA achieves an accuracy exceeding 90% for 28 out of the 45 classes. We note that of all the categories, palaces have the lowest classification accuracy of 32% and are mainly misclassified as commercial areas and churches. This is attributed to their highly similar appearance in remote sensing images, which poses a challenge to our approach. However, we believe that this is reasonable to some extent since these categories can all be classified as buildings.

4) Comparison of Model Parameters and Running Efficiency: Table IV illustrates the comparison of model parameters and running efficiency between MLLA and other SSL methods for the NWPU-RESISC45 dataset. For the model parameters, our method has 3.2 M additional parameters in the training process compared to other methods. This is mainly due to the inclusion of two auxiliary classification heads in our method. Note that, these additional parameters do not affect the model during testing, since we discard the auxiliary classification heads at that stage. For the training time, our method requires a longer training time. This is mainly because our method involves performing k-means clustering for each batch to update the multi-level prototypes efficiently, which consumes a lot of training time. However, in the testing phase, the inference throughput of MLLA is the equal to other methods because the k-means clustering is omitted. In future work, we will explore more efficient clustering methods that can further reduce the training time while ensuring efficient update of multi-level semantic prototypes.

TABLE IV
COMPARISON OF MODEL PARAMETERS AND RUNNING EFFICIENCY

Method	Parmeters (train/test)	Training Time	Testing Time
Supervised	21.3M/21.3M	7min	11s
Pseudo-Label	21.3M/21.3M	38min	11s
Mixmatch	21.3M/21.3M	1h38min	11s
UDA	21.3M/21.3M	47min	11s
VAT	21.3M/21.3M	2h27min	11s
Fixmatch	21.3M/21.3M	51min	11s
Flexmatch	21.3M/21.3M	51min	11s
Dash	21.3M/21.3M	47min	11s
Softmatch	21.3M/21.3M	52min	11s
Freematch	21.3M/21.3M	53min	11s
MLLA	24.5M/21.3M	2h19min	11s

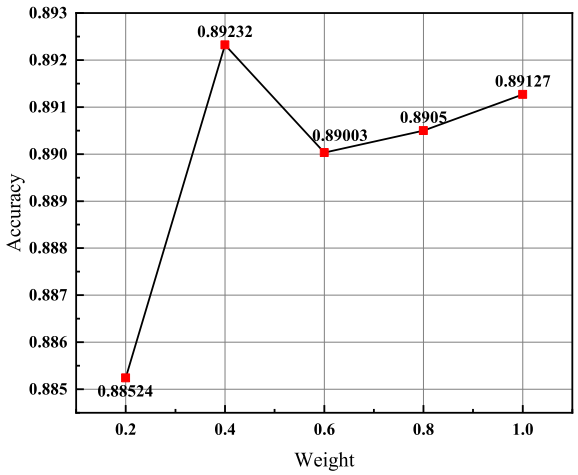


Fig. 7. Effect of the multi-level unsupervised loss weight on accuracy on the NWPU-RESISC45 dataset.

D. Ablation Experiments

We conduct ablation studies to gain a better understanding of the effectiveness of our kernel design. All ablation experiments are performed on the representative NWPU-RESISC45 dataset, which contains a larger number of categories and data. All models are trained for 200 epochs to ensure a fair comparison.

1) Effects of Multi-Level Unsupervised Loss Weight λ: In Section III-B, we follow the common practice of setting the unsupervised loss weight λ to 1 to be fair compared to other methods. Here, we further investigate the impact of λ on the classification accuracy by setting λ to 0.2, 0.4, 0.6, 0.8, and 1, respectively. The results are presented in Fig. 7. When λ increase from 0.2 to 0.4, MLLA achieves the best classification accuracy. However, as the weight continues to increase, the performance of MLLA starts to slightly decline. This phenomenon is probably because if the unsupervised loss weight is too low, the multi-level pseudo-labels of MLLA

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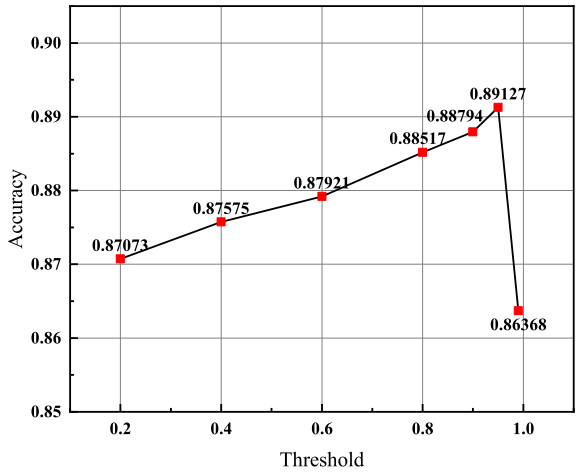


Fig. 8. Effect of the threshold confidence on accuracy on the NWPU-RESISC45 dataset.

is not fully utilized, and vice versa, if the unsupervised loss weight is too high, the role of the labeled data is weakened.

2) Effects of Confidence Threshold Confidence τ: The confidence threshold τ plays an important role in MLLA, which determines the trade-off between the quantity and quality of pseudo-labels. As shown in Fig. 8, we set the τ to 0.2, 0.4, 0.8, 0.9, 0.95, and 0.99, and report the classification accuracy of MLLA with different τ. As we can observe, with increasing thresholds, the model's accuracy improves steadily due to the elimination of a large number of erroneous pseudo-labels. When the threshold is increased to 0.95, the model reaches the highest accuracy, after which it decreases rapidly. The main reason for this is that the excessively high threshold limits the number of samples with pseudo-labels, which leads to underutilization of unlabeled data.

3) Effects of Class Levels and the Number of Classes at Different Levels: The proposed MLLA requires only single-level labels, and the multi-level semantic structure is captured through unsupervised clustering. It is necessary to determine the category levels and the number of classes in advance. To evaluate the effect of different class levels and number of classes, we compare the classification accuracy under various conditions. Specifically, the number of classes at each level is set to 1/2, 1/3, 1/4, 2/3, and 3/4 of the previous level, and the category levels are 2, 3, as shown in Table V. We observe that the model achieves the best performance when the category level is set to 3 and the number of prototypes is set to 45-22-11. Such a prototype setting can well model the multi-level semantic structure in remote sensing images. In the case that the number of prototypes at different levels is too high or too low, such as setting the prototype as 44-11-2 or 44-33-24, the model performance is slightly degraded. This may be because such prototype settings do not correspond to the multi-level semantic structure in remote sensing images, which also demonstrates the effectiveness of the multi-level semantic structure.

TABLE V
ABLATION EXPERIMENTS ON THE NUMBER OF PROTOTYPES AND THE NUMBER OF SEMANTIC LEVELS

Levels	Number of Prototypes	Accuracy
2	45-11	88.85±0.43
	45-15	89.08±0.35
	45-22	88.76±0.39
	45-30	88.90±0.28
	45-33	88.75±0.87
3	45-11-2	88.52±1.02
	45-15-5	88.71±1.02
	45-22-11	**89.13±0.61**
	45-30-10	88.97±0.63
	45-33-24	88.64±0.81

E. Visualization Results

For a more intuitive understanding of our approach, we show visualizations of the high-dimensional features and prediction results obtained by supervised learning, Fixmatch, and the proposed MLLA. These methods are trained using ten training samples per class on the NWPU-RESISC45 dataset.

1) Visualization of Deep Feature Representation: In Fig. 9, we employ t-SNE [67] to reduce the high-dimensional features to 2-D and plot the feature distribution of the NWPU-RESISC45 validation set. As we can see, the feature points of different classes are highly intermixed in the feature distribution generated by supervised methods using only labeled data. However, in the feature distributions generated by FixMatch and the proposed MLLA, smaller intra-class distances are observed, and the features of different classes are more distinguishable. Compared to Fixmatch, there are fewer stray scatter points in the proposed MLLA. This indicates that MLLA learns more discriminative features using multi-level pseudo-labels.

2) Visualization of Prediction Result: Class activation map (CAM) [68] is a widely used method for visualizing model predictions, which can show the importance of different regions in an image for the final prediction. As shown in Fig. 10, we employ CAM to generate response maps for different scene images in the NWPU-RESISC45 validation set, where the regions with more intense red color indicate more importance for the prediction. Compared with supervised methods and Fixmatch, the proposed MLLA focuses on a more complete region where the classification target is located.

V. DISCUSSION

Recently, SSL has achieved remarkable success in reducing the dependence on labeled data. However, existing semi-supervised classification methods ignore the multi-level semantic structure that is prevalent in remote sensing images. The proposed MLLA focuses on capturing the multi-level semantic structure in unlabeled data to further improve the scene classification performance with limited labeled samples. Specifically, we proposed a multi-level prototype awareness module to capture the multi-level semantic structure underlying the remote sensing images. Furthermore, based on the multi-level semantic structure, we design a multi-level pseudo-label generation module to assign multi-level pseudo-labels to unlabeled remote sensing images. Finally, the classification model is progressively trained by combining the labeled

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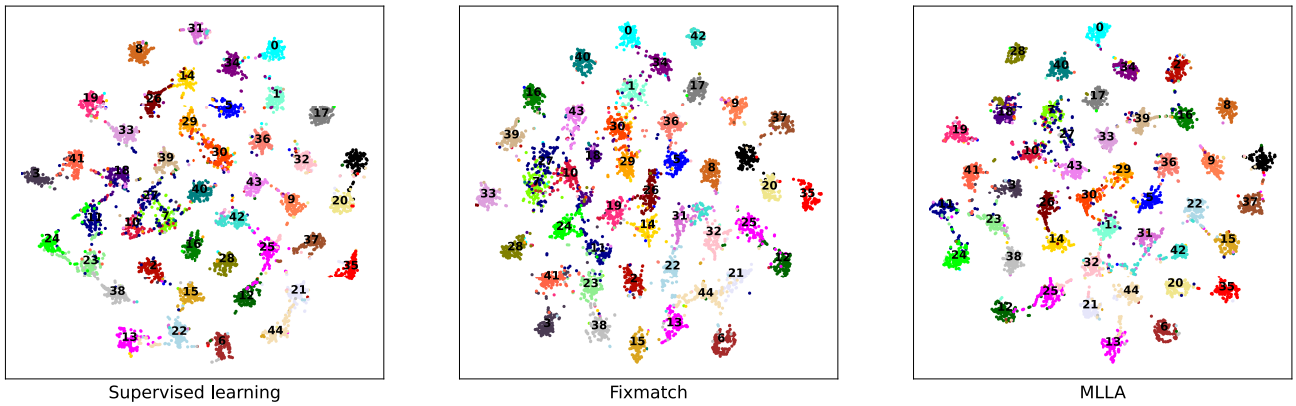


Fig. 9. Visualization of high-dimensional features learned by supervised learning, Fixmatch and the proposed MLLA, respectively, using t-SNE.

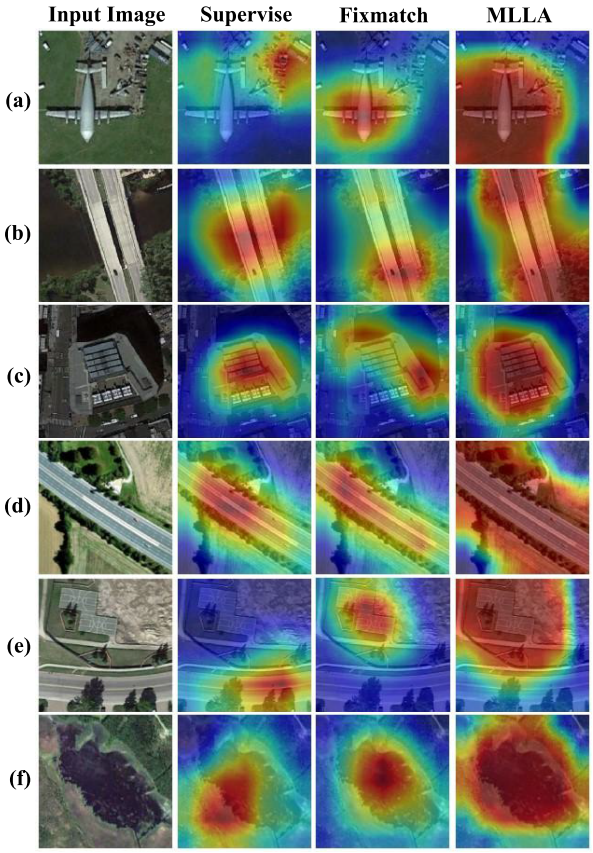


Fig. 10. Response maps of the supervised method, Fixmatch, and the proposed MLLA on different scene images. (a) Airplane. (b) Bridge. (c) Commercial area. (d) Freeway. (e) Basketball court. (f) Lake.

images and the multi-level pseudo-labeled images. Quantitative results in Section IV on three popular remote sensing datasets demonstrate that the proposed MLLA outperforms the state-of-the-art SSL methods. By capturing the multi-level semantic structure, MLLA shows great potential for remote sensing scene classification tasks with limited labeled samples. Moreover, an attractive highlight of MLLA is its ability to capture the multi-level semantic structure in remote sensing images with only single-level annotations, which ensures its availability in most remote sensing datasets.

As observed in the confusion matrices of the three datasets in Figs. 3–5, MLLA tends to make misclassifications on some highly similar-looking classes, such as dense versus moderate housing, rivers versus beaches, highways versus Runways, and palace versus church. One possible reason for this is that the differences in these scene classes are mainly in local details and current classification methods that rely on global features do not effectively capture the differences that exist in these images. Therefore, the initial pseudo-labels may have been assigned incorrectly, leading to an error accumulation. In future work, we plan to use more fine-grained, multi-level local features to further distinguish these classes. In addition, the complex background in remote sensing images is also an important source of misclassification. Therefore, how to eliminate the background interference is also one of our next research directions.

VI. Conclusion

In this work, we proposed a MLLA semi-supervised scene classification framework, called MLLA, which effectively exploits the multi-level semantic structure in remote sensing images to improve the scene classification performance. The framework consists of two core modules: a multi-level prototype awareness module and a multi-level pseudo-label module. First, the multi-level prototype awareness module captures the multi-level semantic structure in remote sensing images. Then, based on this structure, the multi-level pseudo-label generation module assigns multi-level pseudo-labels to unlabeled remote sensing images to train the classification model. Extensive experiments show that the proposed MLLA achieves improved performance over the state-of-the-art methods in semi-supervised remote sensing scene classification tasks.

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